

Assessment of a new CAD-generated imaging marker to predict risk of having mammography-occult tumors

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ABSTRACT

Since sensitivity of mammography is limited in detecting subtle cancer, we propose to develop a new computer-aided detection (CAD) scheme to generate a new quantitative image marker to predict risk of having mammography-occult cancer that can be detected by breast MRI. The study is based on the hypothesis that overall breast density and bilateral asymmetry of breast density between left and right breasts associate with higher risk of developing breast cancer in short-term. Thus, a new CAD scheme is developed to process images and analyze bilateral asymmetry of mammographic density and tissue structure. From the computed image features, a machine learning model is trained to generate an image marker or likelihood score to predict risk of having mammography-occult tumors. In this presentation, we report two cases in which screening mammograms are rated as BIRADS 2 by radiologists. Two women do not qualify for breast MRI screening due to the lower risk scores predicted by existing epidemiology risk models. CAD scheme analyzes mammograms of these two cases and produces high risk scores of having mammography-occult tumors. After applying breast MRI screening, two mammography-occult tumors are detected. Biopsy results confirm one invasive ductal carcinoma (grade 3) and one high-risk tumor of solitary breast papillomas, which needs to be removed by surgery. This study demonstrates potential advantages of applying CAD-generated image marker to detect abnormality or predict cancer risk that are missed or overlooked by radiologists. It can thus increase efficacy of using MRI as an adjunct tool to mammography to detect more subtle cancers.

Keywords: Breast cancer screening, Mammography-occult cancer, Quantitative image marker, Breast magnetic resonance imaging (MRI).

I. INTRODUCTION

Breast cancer is the most prevalent cancer and the second leading cause of cancer death in women [1]. Scientific evidence has shown that earlier cancer detection and treatment reduced patients' mortality rates [2]. Mammography is the most popular and widely accepted imaging modality in population-based breast cancer screening to date. However, due to the large variability of breast abnormalities and overlapped dense fibro-glandular tissue in mammograms, mammography performance is not satisfactory in both cancer detection sensitivity and specificity [3]. Detection sensitivity of screening mammography decreases among the women who are younger age, have dense breasts, use hormone replacement therapy, and carry certain breast cancer susceptibility genes. For example, one study reported that mammography sensitivity reduced from 87.0% in women with almost entirely fatty breasts to 62.9% in women with extremely dense breasts or reduced from 83.3% in women over 80 years old to 68.6% in women younger than 50 years old [4]. As a result, a high percentage of mammography-occultly subtle breast tumors or cancer are missed or overlooked by radiologists in reading and interpreting screening mammograms.

In order to help detect more early cancers in the breast cancer screening environment, many advanced medical imaging modalities have been developed and investigated. Studies have found that using a multi-modality imaging screening approach has potential to significantly increase cancer detection sensitivity. Among other imaging modalities, dynamic contrast-enhanced breast magnetic resonance imaging (MRI) has the highest cancer detection sensitivity as demonstrated in several large clinical trials. For example, a previous ACRIN multi-institutional study investigated and compared benefit

of detecting invasive breast cancers with addition of annual screening ultrasound or a single screening MRI to mammography in 2,809 women with elevated breast cancer risk. The study reported that only 53.2% (59 of 111) cancers were detected by mammography alone. Among the mammography-occult cancers, using ultrasound and breast MRI detected 32 and 41 cancers, respectively [5]. As a result, breast MRI has been recommended by American Cancer Society as an adjunct to mammography for women at an elevated breast cancer risk (e.g. > 20–25% of lifetime risk) [6]. However, since majority of breast cancers are detected in women without any known high-risk factors [7], these women are excluded from breast MRI screening. Thus, how to more accurately identify women who have higher risk of having mammography-occult tumors or cancer in the general breast cancer screening population and/or environment remains an unsolved clinical challenge or issue to date.

In order to better address this clinical challenge, we in the recent studies have developed and tested a new and novel computer-aided detection (CAD) scheme that processes and analyzes the negative mammograms acquired in routine breast cancer screening, which aims to explore and identify new quantitative mammographic image markers based on bilateral asymmetry of breast density and tissue structure between the left and right breasts in order to more accurately predict the short-term breast cancer risk (i.e., the risk of having mammography-detectable breast cancer in the next sequential mammography screening) [8]. Our previous studies have demonstrated that using this CAD-generated mammographic image marker could yield significantly higher discriminatory power (or odds ratios) in predicting risk of having or developing mammography-detectable cancer in the next sequential screening after current negative mammography screening than using other existing breast cancer risk factors [9] and an association or increasing trend between the increased prediction scores of the image marker and the increased risk of having mammography-detectable cancer [10]. Therefore, based on these promising study results and continuous optimization of this quantitative image marker [11], we hypothesized that this quantitative mammographic image marker might also have potential to help predict the risk of having mammography-occult tumors or cancer. As a result, using breast MRI modality has potential to help detect more subtle tumors or cancer among the women who do not qualify for breast MRI screening based on current clinical guidelines. In this paper, we report preliminary findings of our study that applying this new quantitative image marker to predict risk of having mammography-occult cancer, which can be detected by breast MRI.

II. MATERIALS AND METHODS

In predicting breast cancer risk, breast density assessed from mammograms has been well recognized as one of the most prevalent risk factors of developing breast cancer in long-term or lifetime of the women, which has attracted broad research interest [12]. However, we also learned that the bilateral asymmetry of breast density or tissue structure is also an important radiographical phenotype related to biological processes associated with abnormality and radiologists routinely examine bilateral mammographic tissue asymmetry and its change over time in their decision-making process to detect subtle breast cancer. In order to implement above concept into a new quantitative mammographic image marker to predict short-term risk of having or developing breast cancer after negative mammography screening, we developed a new CAD scheme that automatically detects and quantitatively assesses bilateral asymmetry of mammographic density or tissue structure between left and right breasts. The details of this new type of CAD scheme have been reported elsewhere in our previous studies [8-11].

In brief, when using this new CAD scheme, two bilateral mammograms of left and right breasts are uploaded simultaneously into the specially designed CAD platform or graphic user interface (GUI) as shown in Figure 1. For two bilateral cranio-caudal (CC) view mammograms, CAD scheme directly applies image processing filters to generate several filtered maps, while for two bilateral medio-lateral oblique (MLO) views of mammograms, CAD scheme will first segment and remove pectoral muscle regions from the mammograms [13] before applying image filters to process images. As shown in Figure 1, four filtered maps that represent (a) distribution of dense fibro-glandular tissue, (b) tissue local pixel value variation, (c) the segmented focal regions and (d) Gaussian bandpass filtering results are computed. Then, from these filtered image maps, CAD scheme computes bilateral differences of several statistical features including the segmented region size, the mean pixel values, standard deviation and skewness. Using the selected image features, a pretrained multi-feature fusion based artificial neural network is implemented into the CAD scheme to predict short-term breast cancer risk (i.e., the likelihood of women having or developing breast cancer that can be detected in the next sequential breast cancer screening). For example, in one study involving 690 study cases, the adjusted odds ratio shows an increased risk trend with the increased prediction scores from 1.0 to 9.07 between the positive and negative screening cases in the next sequential mammography screening [14].

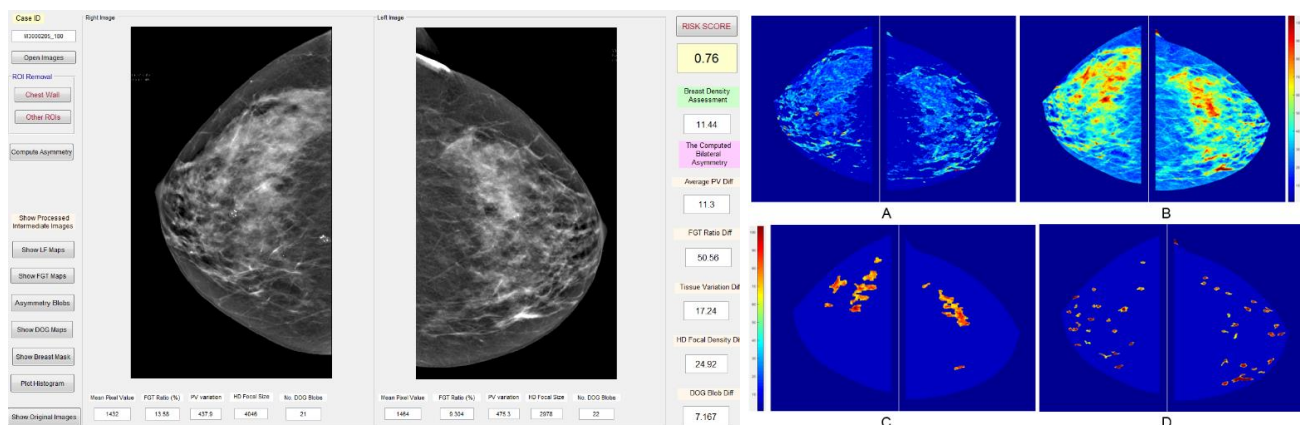


Figure 1: (Left) Illustration of GUI of our CAD scheme to detect short-term breast cancer risk (*i.e.*, in example the risk score is 0.76 or 76%). (Right) Showing intermediate results of image processing steps including the bilateral local density fluctuation map (A), the computed breast density map (B), detected focal density regions (C) and Gaussian bandpass filtering generated map (D). Color bar indicates volumetric density levels of the pixel values.

Recently, we started to investigate and test the feasibility of applying this new CAD-generated quantitative image marker to predict the risk of having mammography-occult tumors or cancer that can be detected by breast MRI. Under the approval of IRB of Mercy Health System, we recruited women who participated in regular mammography-based breast cancer screening in Breast Center of Mercy Hospital in Oklahoma City and have negative detection and diagnostic results namely, BIRADS 1 or 2 rated by radiologists in reading and interpreting these mammograms. These negative screening mammograms are then processed and analyzed by our CAD scheme. If the CAD-generated average risk score computed from two pairs of bilateral mammograms of both CC and MLO views is greater than a pre-determined threshold (*i.e.*, $S \geq 0.8$), the women are recommended to take a breast MRI screening as an adjunct to mammography screening. In this paper, we report two study cases and discuss our preliminary findings.

III. RESULTS

3.1. Case One

This woman is 67 years old. Mammographic breast density was originally rated as D by the radiologist indicating that the breasts are extremely dense (>75% glandular), which lowers the sensitivity of mammography. The screening mammograms are read and rated by radiologist. The final diagnostic rating is BIRADS 2 (benign and no suspicious findings for cancer). The woman has no family history of breast and/or ovarian cancer. The 10-year and lifetime breast cancer risks are 8.2% and 13.7%, respectively. Thus, this woman does not qualify to take a breast MRI based on current breast cancer screening guidelines.

Four screening mammograms are processed and analyzed using our CAD scheme. As shown in Figure 2, the CAD-computed risk scores are 0.73 and 0.91 from bilateral CC view and MLO view images, respectively. The average risk score of this case is 0.82, which is greater than the risk threshold. Thus, the woman is recommended to taken breast MRI screening. A mammography-occult tumor is visually detected on MRI (as shown in Figure 2(c)). The tumor has a diameter of 1.6cm measured on MRI. The biopsy result of this tumor confirms that this is an invasive ductal carcinoma (IDC) rated as grade 3, which is poorly differentiated and probably grow and spread faster.

3.2. Case Two

This woman is 60 years old. The mammographic breast density was originally rated as C indicating that breasts are heterogeneously dense (approximately 51% – 75% glandular), which may obscure small masses. The screening mammograms are rated by radiologist as BIRADS 2. The woman has family history of breast cancer. Her grandmother had history of breast cancer. The 10-year and lifetime breast cancer risks are 10.9% and 25.8%, respectively.

Four screening mammograms of this case are also processed and analyzed by our new cancer risk prediction-based CAD scheme. As shown in Figure 3, the CAD-computed risk scores are 0.85 and 0.76 from bilateral CC view and MLO

view images, respectively. Thus, the average risk score of this case is 0.805, which is also greater than the risk threshold. After taking breast MRI screening, a mammography-occult suspicious tumor is visually detected on MRI (as shown in Figure 3(c)). The biopsy result confirms that this is high-risk solitary breast papillomas with atypical ductal hyperplasia (ADH) and atypical lobular hyperplasia (ALH). Based on current clinical guidelines, surgical excision is then performed to remove the tumor.

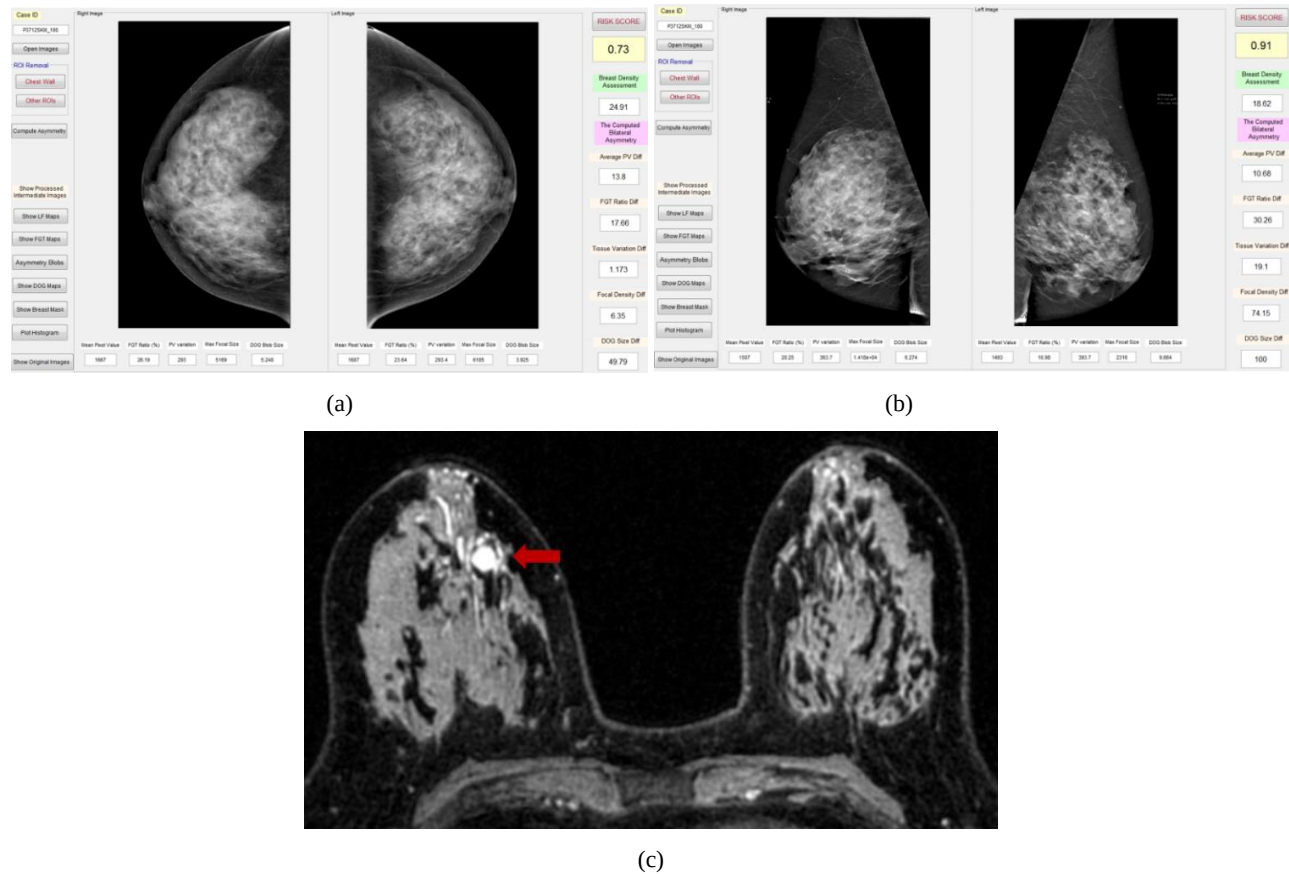


Figure 2: Two bilateral CC view images (a) and two bilateral MLO view images (b) of case 1 displayed on CAD platform and the computed risk scores, and (c) one MRI slice showing a suspicious tumor (as pointed by a Red arrow).

IV. DISCUSSION

In this paper, we reported and analyzed above two study cases, which provide the preliminary observations or findings for a new technology assessment. To the best of our knowledge, no similar study observations have not previously reported in the literature. This new study along with our observations have following unique characteristics. First, although it is well known that detecting subtle and/or early breast cancer in reading mammograms is often a difficult task for radiologists because many subtle tumors may occult by heterogeneous and dense fibro-glandular tissue overlapping, mammograms contain useful and discriminatory image features that can be used to assist cancer detection. In recent medical image informatics field, radiomics concept has been attracted broad research interest, which indicates that tumor phenotypes can be comprehensively quantified by applying large number of quantitative image features [15]. As a result, many studies have been reported in the literature to classify between malignant and benign tumors [16], predict different types of tumors or tumor stages (i.e., histologic grade of invasive breast cancer using mammograms [17] and Fuhrman grades of renal clear cell carcinoma using CT images [18]), risk of tumor metastasis [19] and tumor prognosis [20]. In this study, we demonstrate a new application field of using radiomics concept and the feasibility of identifying and applying novel quantitative mammographic image features to predict risk of screening negative mammograms depicting the mammography-occult tumors or cancer.

Second, although high risk clinics have been established in many medical centers or hospitals to analyze and predict breast cancer risk of women using the epidemiology study-based breast cancer risk models and thus recruit high-risk women by adding breast MRI as an adjunct imaging screening to mammography, our study show that quantitative risk scores computed from bilateral mammograms are different or do not highly associate from the risk prediction scores assessed using the existing epidemiology study-based breast cancer risk models. This is encouraging and indicates that applying this new quantitative mammographic image marker or risk score can identify a different group of women who are now excluded from breast MRI screening based on current clinical guidelines. Thus, using or adding this new risk prediction marker or method has potential to help detect more subtle cancers at an early stage in the population-based breast cancer screening practice.

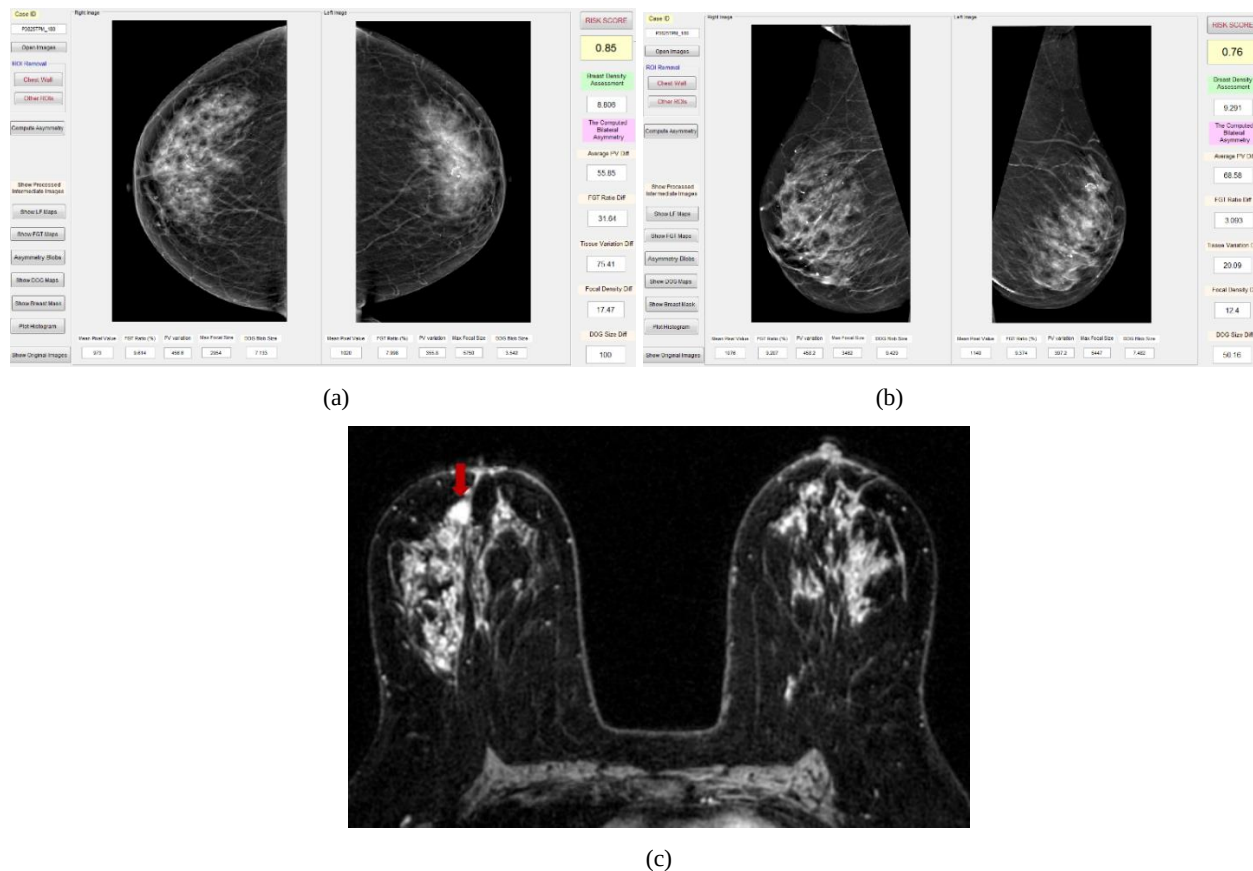


Figure 3: Two bilateral CC view images (a) and two bilateral MLO view images (b) of case 2 displayed on CAD platform and the computed risk scores, and (c) one MRI slice showing a suspicious tumor (as pointed by a Red arrow).

Third, the commercialized CAD schemes of mammograms are available and routinely used in the current clinical practice to assist radiologists in detecting mammography-visible breast tumors. Our study aims to expand the concept of CAD schemes of mammograms to a new application field, which aims to help predict risk of the negative screening mammograms depicting the mammography-occult tumors or cancer. This is a new type of CAD scheme to explore a new set of optimally relevant radiomics features, which is different from previous CAD approach to identify quantitative image features to detect visible suspicious tumors [21] or use results of the existing CAD schemes to predict short-term breast cancer risk [22].

We also recognize that this is just a preliminary case report type study. Thus, although the study demonstrates feasibility of developing such a new type of CAD scheme, more research and development efforts are needed in the future studies, which include to (1) develop and test more effective image processing algorithms to segment and generate mammographic density and tissue structure maps including using deep learning model based image segmentation methods [23], (2) compute and select more clinically-relevant and non-redundant radiomics features associated with cancer risk

[24], (3) train and validate the optimal multi-feature fusion-based machine learning models, and (4) assemble the large and diverse image datasets by collecting more study cases with the confirmed mammography-occult tumors or cancer in different medical institutions.

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